Table 5.3: Complex mixture

S.No	Composition	Approximate percentage
1.	Liquid (Water)	87%
2.	Solids	13%
3.	Fat	4%
4.	Protein	3.3%
5.	Lactose (Milk Sugar)	5%
6.	Ash content (Vitamins and minerals)	0.7%

its immediate surroundings, and the milking and milk handling equipment.

- Bacteria that get on to the outside of the teat may be able to invade the opening and hence the udder interior. The organisms most commonly isolated are micrococcus, streptococci and the diptheroid *Corynebacterium bovis*. Aseptically taken milk from a healthy cow normally contains low number of organisms, typically fewer than 10^2–10^3 cfu ml^{-1}
- The udder exterior and its immediate environment can be contaminated with organisms from the cow's general environment. Heavily contaminated teats have been reported to contribute up to 10^5 cfu ml^{-1} in the milk. Contamination from bedding and manure can be source of human pathogens such as *E.Coli, Campylobacter, Salmonella, Bacillus spp.* and *Clostridia spp.*
- Milk – handling equipment such as teat cups, pipe work, milk holders and storage tanks is the principal source of the microorganisms found in raw milk. *Micrococcus* and *Enterococcus*.

5.5.4 Microbiological Standard and Grading of Milk

In India, raw milk is graded by Bureau of Indian standards (BIS) 1977. The Indian standard institute (ISI) has prescribed microbiological standard for quality of milk.

- Coliforms count in raw milk is satisfactory if, coliforms are absent in 1:100 dilution.
- Coliforms count in pasteurized milk is satisfactory is coliforms are absent in 1: 10 dilution (Table 5.4).

Grading of milk

The quality of milk is judged by certain standards and it is known as grading milk. Grading of milk is based upon regulations pertaining to production, processing and

Table 5.4: Microbiological Standard and Grading of Milk

S.No	Product	Temperature	Bacterial count/ml	Chemical and others
1.	Grade A raw milk for pasteurization	Cooled to 50°F and maintained there at until processed	Individual producer milk should not exceed 100,000/ml prior to combining with other produce of milk	Antibiotics should be less than 0.05 unit/ml
2.	Grade A pasteurized milk products	Cooled to 45°F or less	Milk and Milk products 20,000/ml coliforms limit not exceeding 10/ml	Phosphates less than 1mg/ml

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